# aster

Processed on: 2025-11-07 11:04:37

ASP version: 3.6.0-alpha

asp_plot version: 3.0.0

## DEM Summary

| Property | Value |
| --- | --- |
| DEM File | run-DEM.tif |
| Dimensions (px) | 4348 x 4128 |
| GSD (m) | 17.50 |
| CRS | EPSG:32610 |
| Nodata (%) | 30.3 |
| Elevation Range (m) | -15.3 to 4345.4 |
| Acquisition Date | 2017-07-31 19:07:28 UTC |

# Processing Parameters

## Runtime Summary

| Step | Runtime |
|---|---|
| Bundle Adjust | N/A |
| Stereo | 0 hours and 12 minutes |
| point2dem | 0 hours and 0 minutes |

### Bundle Adjust Command:

```
Bundle adjustment not run
```

### Stereo Command:

```
stereo -t aster --stereo-algorithm asp_mgm --subpixel-mode 9 --aster-use-csm out-Band3N.tif out-Band3B.tif out-Band3N.xml out-Band3B.xml out_stereo/run --corr-seed-mode 1 --sgm-collar-size 256 --compute-point-cloud-center-only --threads 8
```

### point2dem Command:

```
point2dem -r earth --auto-proj-center --tr 200 out_stereo/run-PC.tif -o out_stereo/run-200m
```

### Report Generation Command:

```
asp_report --directory /Users/ben/Desktop/asp-plot-examples/aster/ --stereo-directory out_stereo/ --dem-filename run-DEM.tif --subset-km 5.0 --reuse-selections /Users/ben/Desktop/uw-github/asp_plot/reports/ASTER-asp-report_figure_selections.yml --report-filename /Users/ben/Desktop/uw-github/asp_plot/reports/ASTER-asp-report.pdf
```

# Input Scenes

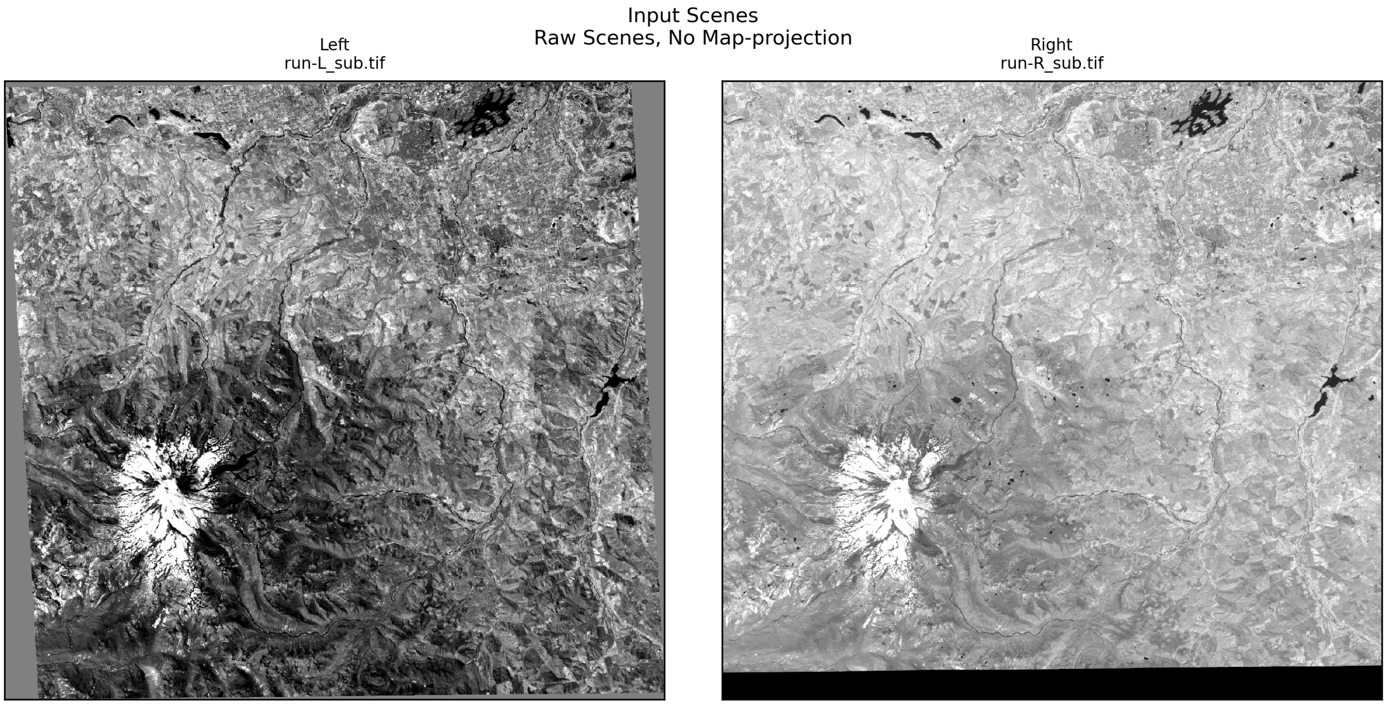


*Figure 1: Left and right input scenes used for stereo processing. Non-mapprojected scenes are shown after ASP's alignment step (e.g., affineepipolar), which rotates images to create horizontal epipolar lines for correlation. Mapprojected scenes have been orthorectified with RPCs against a reference DEM to roughly align the two images prior to correlation, which reduces the disparity search range; they are displayed here in their map-projected orientation.*

## Stereo Geometry

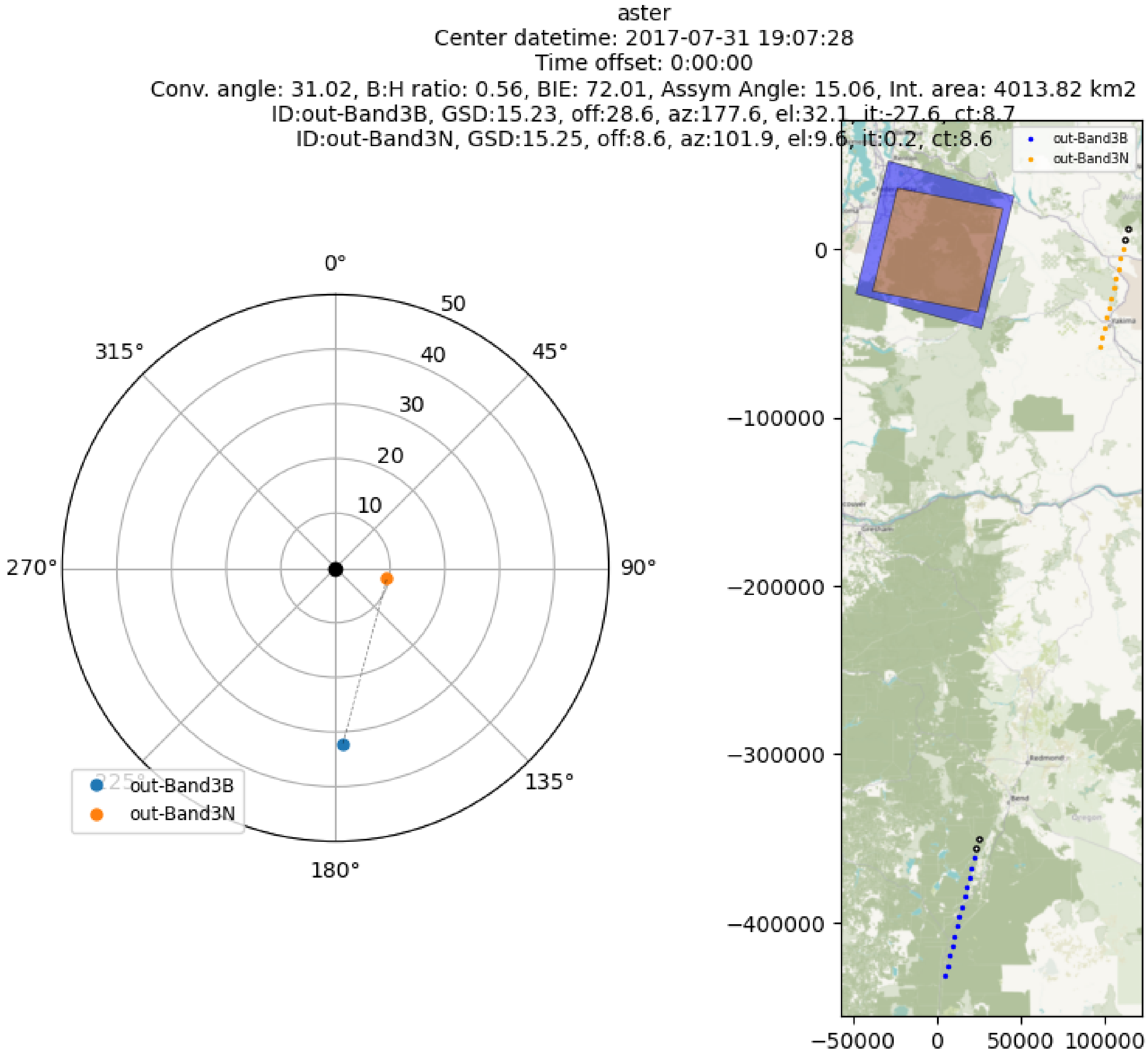


*Figure 2: Stereo acquisition geometry skyplot and map view showing satellite viewing angles and scene footprints.*

## Match Points

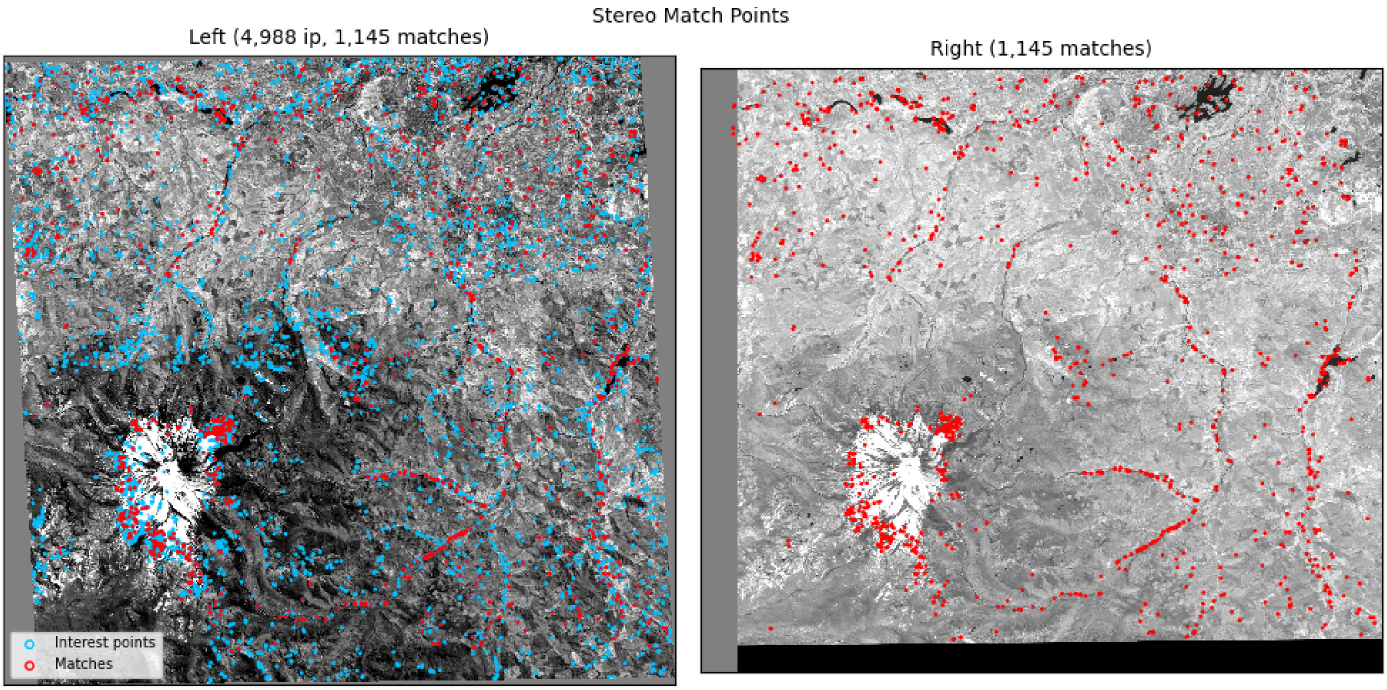


*Figure 3: Interest point matches between left and right images (red), overlaid on the raw interest points detected on each image individually (blue, when the .vwip files are present in the stereo directory). These are produced by stereo_corr during its initial interest point matching step, which is used to set the search windows for subsequent dense correlation (not the dense correlation matches themselves). Comparing the raw interest points against the matches shows whether sparse matching traces back to poor matching or to areas with no detected interest points at all (e.g., limited texture, clouds, water). Some ASP configurations only write a .vwip file for the left image (the right image's interest points are detected on a temporary aligned image and not saved), in which case only matches are shown on the right.*

# Disparity

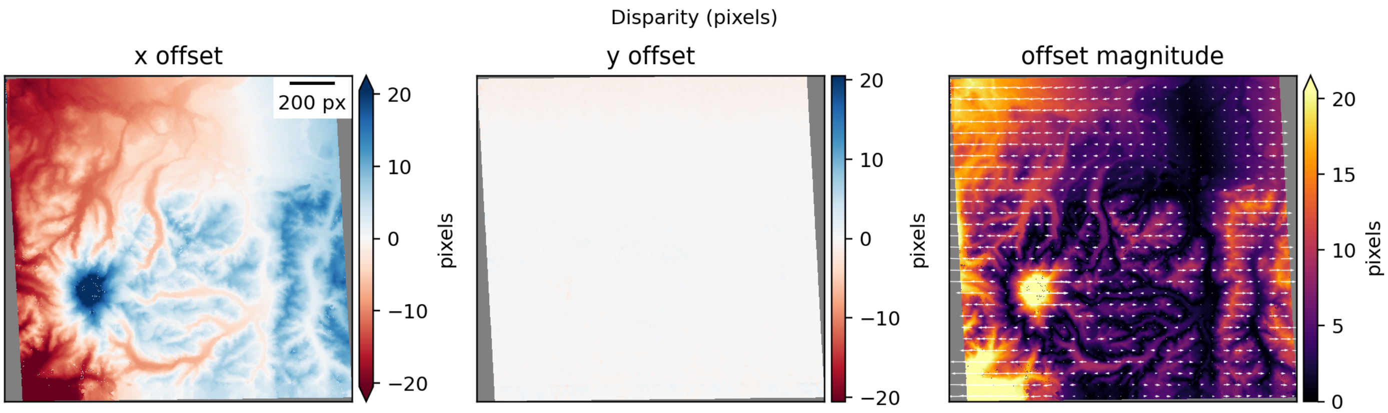


*Figure 4: Horizontal and vertical disparity maps in pixels with quiver overlay.*

## DEM Results

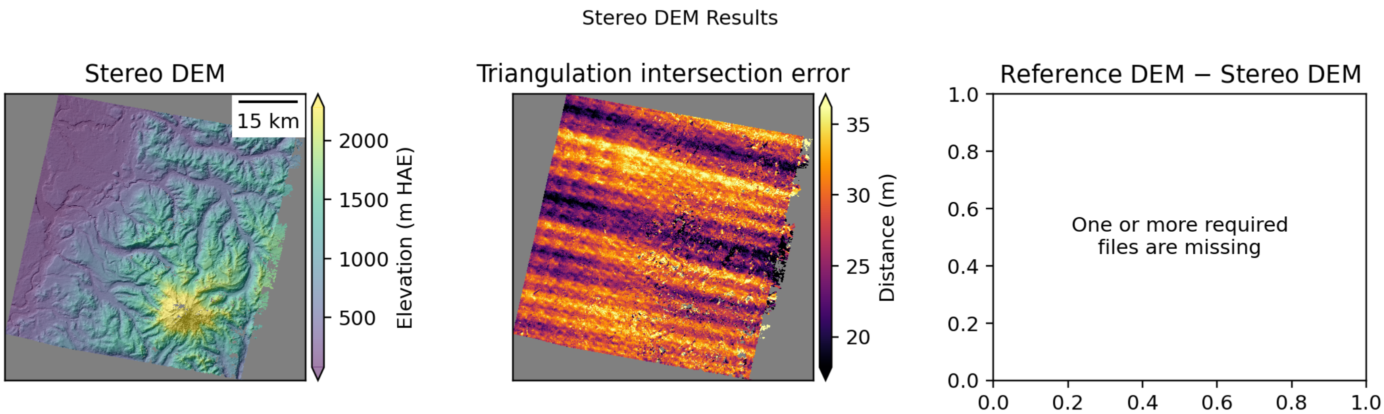


*Figure 5: Output DEM with intersection error map and difference relative to the reference DEM used in processing.*

## Detailed Hillshade

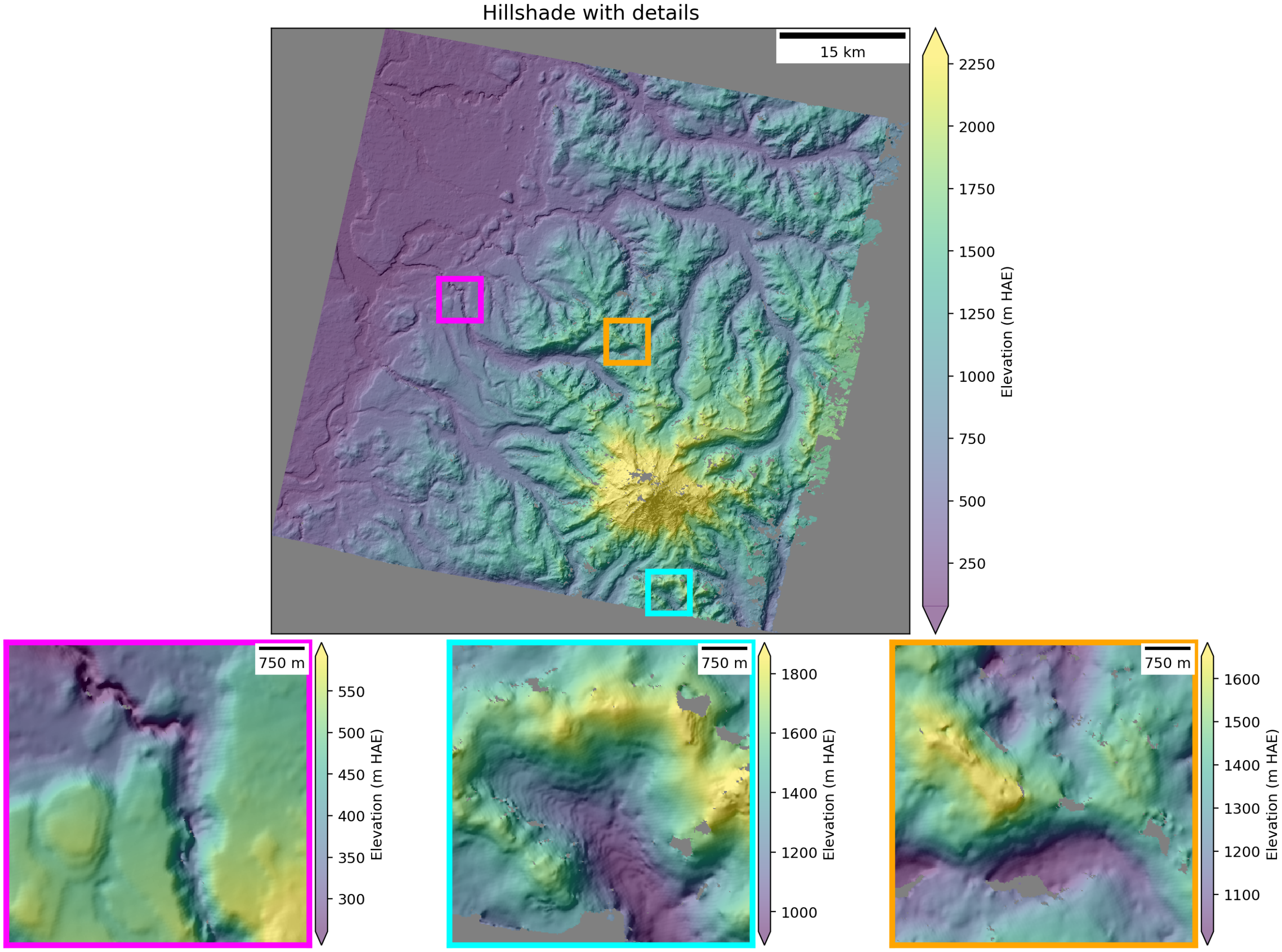


*Figure 6: DEM hillshade. If the intersection error is available, zoomed subsets selected from low, medium, and high (left to right) uncertainty areas are displayed in the second row. If the mapprojected image is available, corresponding ortho image subsets are displayed in the bottom row.*

## ICESat-2 ATL06-SR Map

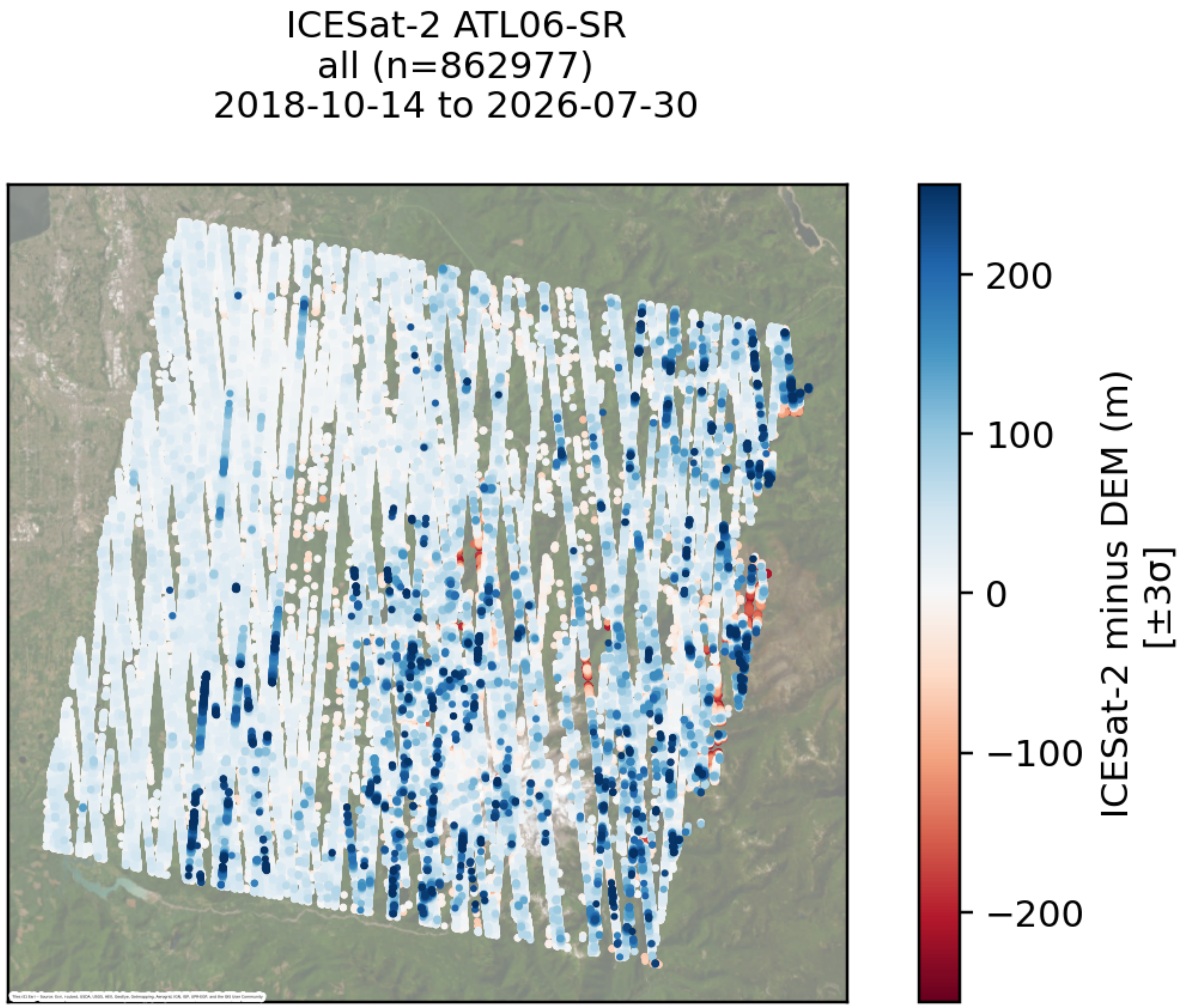


*Figure 7: ICESat-2 ATL06-SR elevation differences vs. ASP DEM.*

# ICESat-2 ATL06-SR Histogram

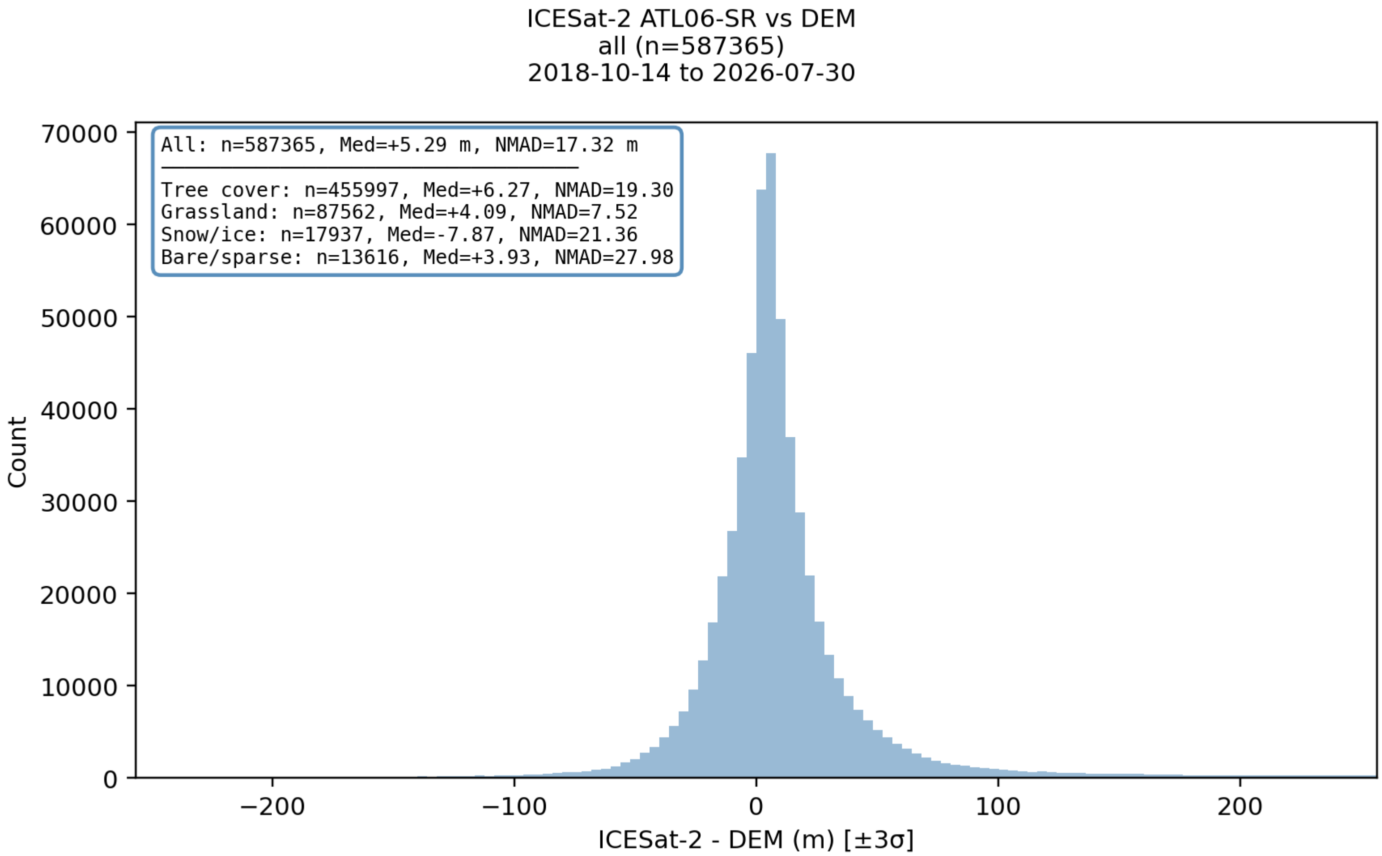


*Figure 8: Distribution of elevation differences between ICESat-2 ATL06-SR and ASP DEM with per-landcover statistics.*

## ICESat-2 ATL06-SR Profile

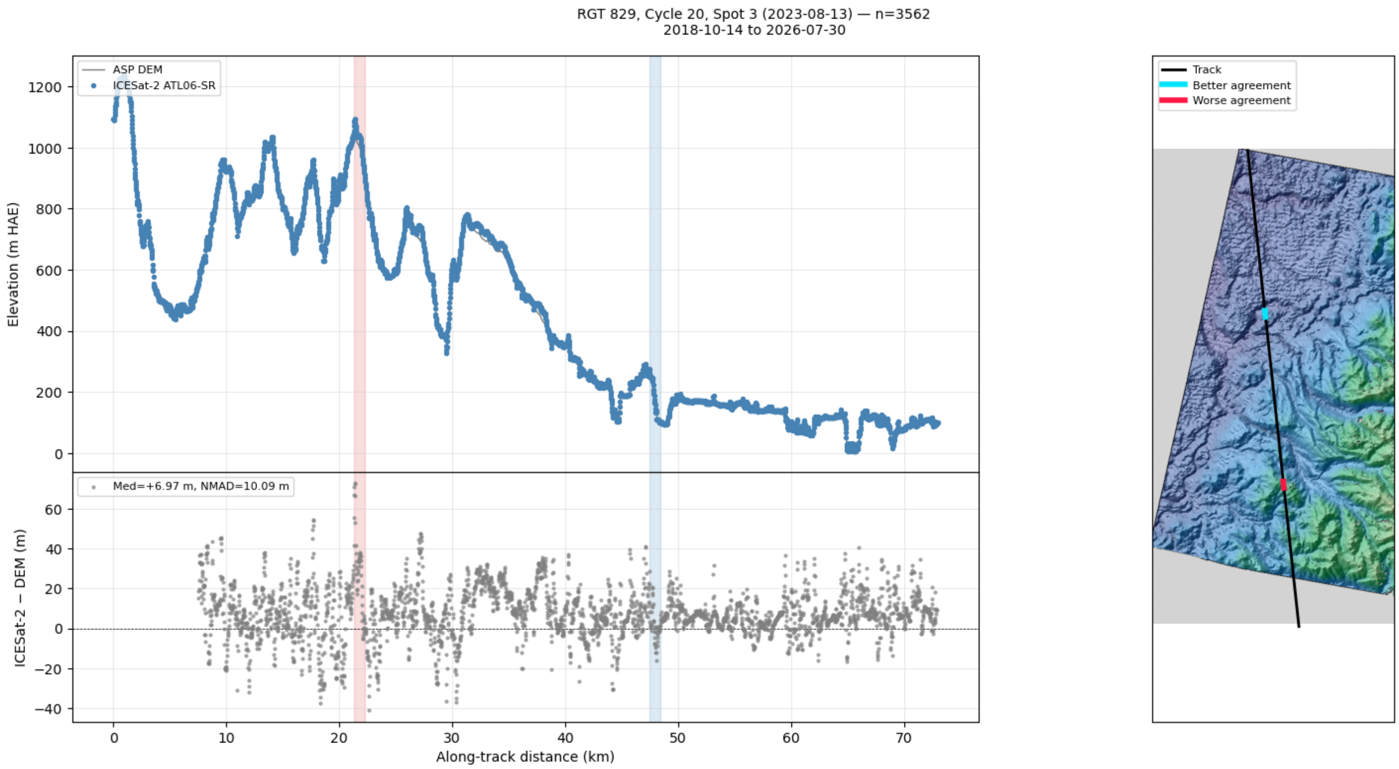


*Figure 9: Elevation profile along the ICESat-2 track with the most valid points, comparing ATL06-SR and DEM heights (top) and height differences (bottom), with a context hillshade map on the right. Blue/red highlights mark the 1 km segments with better and worse agreement between ICESat-2 and the DEM.*

# ICESat-2 ATL06-SR Agreement Segments

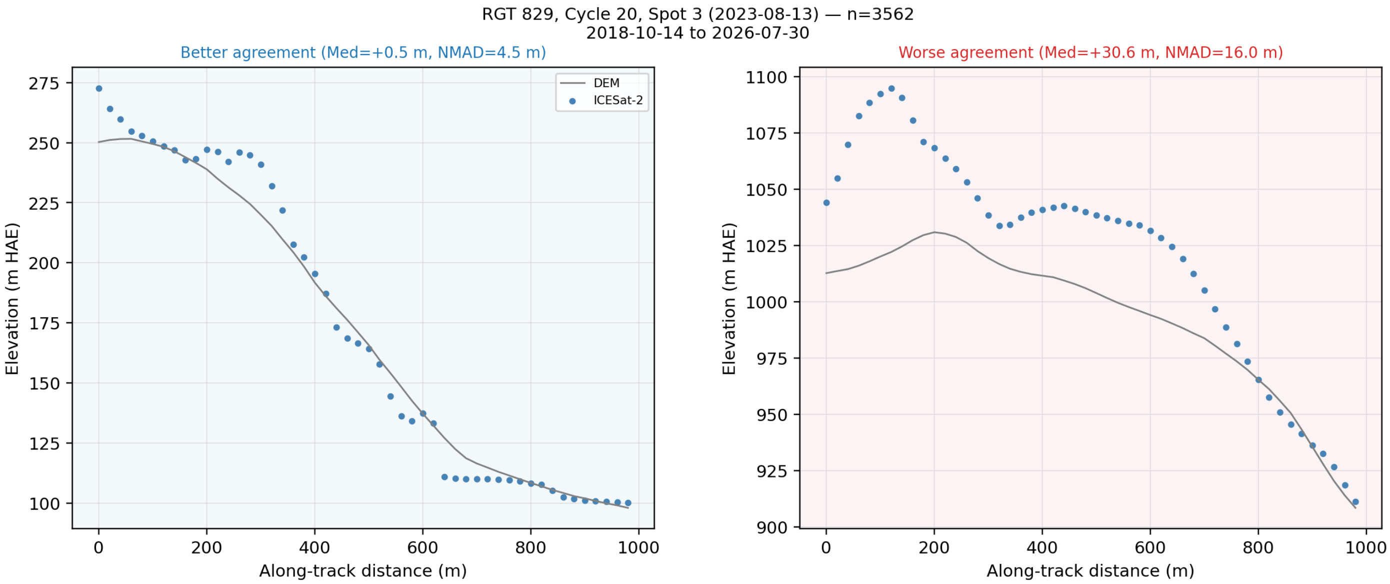


*Figure 10: 1 km segments along the ICESat-2 track with better (left) and worse (right) agreement with the DEM. Segments are scored by 3·|median(dh)| + NMAD(dh), which weights the median bias three times more than the dispersion so that a segment with a large bias is not selected as 'better agreement' just because its NMAD is small.*

# DEM Alignment with ICESat-2

## Alignment Parameters

| Parameter | Value |
|---|---|
| processing_level | all |
| minimum_points | 500 |
| agreement_threshold | 0.25 |
| min_translation_threshold | 0.1 |
| improvement_threshold_pct | 5.0 |

### Error Statistics (m)

| Statistic | Before | After | Change |
|---|---|---|---|
| Median | 7.34 | 6.25 | -14.8% |
| NMAD | 6.30 | 5.88 | -6.7% |
| RMSE | 9.87 | 9.33 | -5.4% |

### Translation (m)

| Component | Value |
|---|---|
| North | 16.2 |
| East | -5.06 |
| Down | 4.25 |
| Magnitude \|T\| | 17.5 |

ASP's pc_align estimates a rigid 3D translation that minimizes the height residuals between the ASP DEM and ICESat-2 ATL06-SR ground-track points used as the reference point cloud. The translation is applied to the DEM directly (geotransform + pixel-value shift, no resampling) to produce the aligned DEM.

Alignment Parameters (above):
- processing_level: ATL06-SR filter key used as the reference; 'all' uses every filtered point.
- minimum_points: minimum ATL06-SR point count required; fewer points skips the alignment.
- agreement_threshold: maximum relative disagreement across temporal sub-filters before the aligned DEM is flagged as inconsistent.
- min_translation_threshold: minimum translation magnitude (as a fraction of the DEM GSD) required to write out an aligned DEM.
- improvement_threshold_pct: minimum percentage reduction in the median residual required to keep the aligned DEM on disk; below this, the aligned DEM is removed.

Error Statistics (above, in meters): each row is one statistic of the DEM-vs-ICESat absolute height residuals, Before and After alignment; Change is the percent change from Before to After, so negative is an improvement.
- Median: median residual.
- NMAD: normalized median absolute deviation; a robust spread, insensitive to outliers.
- RMSE: root-mean-square residual; outlier-sensitive, so a large RMSE next to a small NMAD flags a tail of gross errors rather than a broad misfit.
- p16 / p50 / p84: shown instead of the above when the pc_align log predates ASP 3.7.0 (which added the Median / NMAD / RMSE summary): 16th / 50th / 84th percentile of the residuals; p50 is the median.

Translation (above, in meters): the applied translation vector as North / East / Down components and its magnitude |T|. Positive Down means the DEM was too high.

**Median improved from 7.34 m -> 6.25 m (14.8% reduction). Aligned DEM written to /Users/ben/Desktop/asp-plot-examples/aster/out_stereo/run-DEM_pc_align_translated.tif.**

# ICESat-2 ATL06-SR Histogram (Aligned DEM)

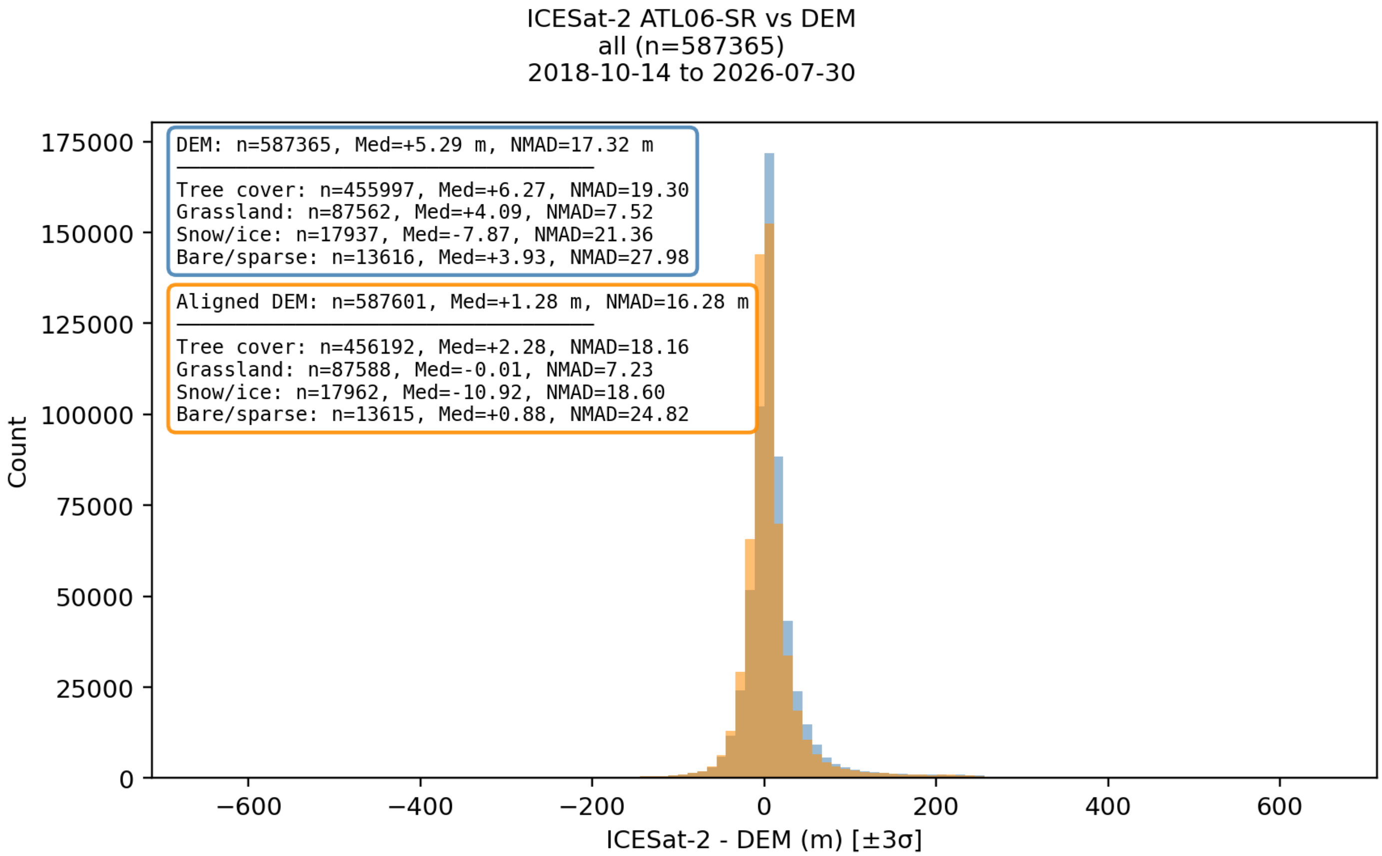


*Figure 12: Pre- (steelblue) and post-alignment (orange) distributions of ICESat-2 minus DEM height differences, with per-landcover statistics in the two stacked text boxes. Box outline color matches the bar color.*

# ICESat-2 ATL06-SR Profile (Aligned DEM)

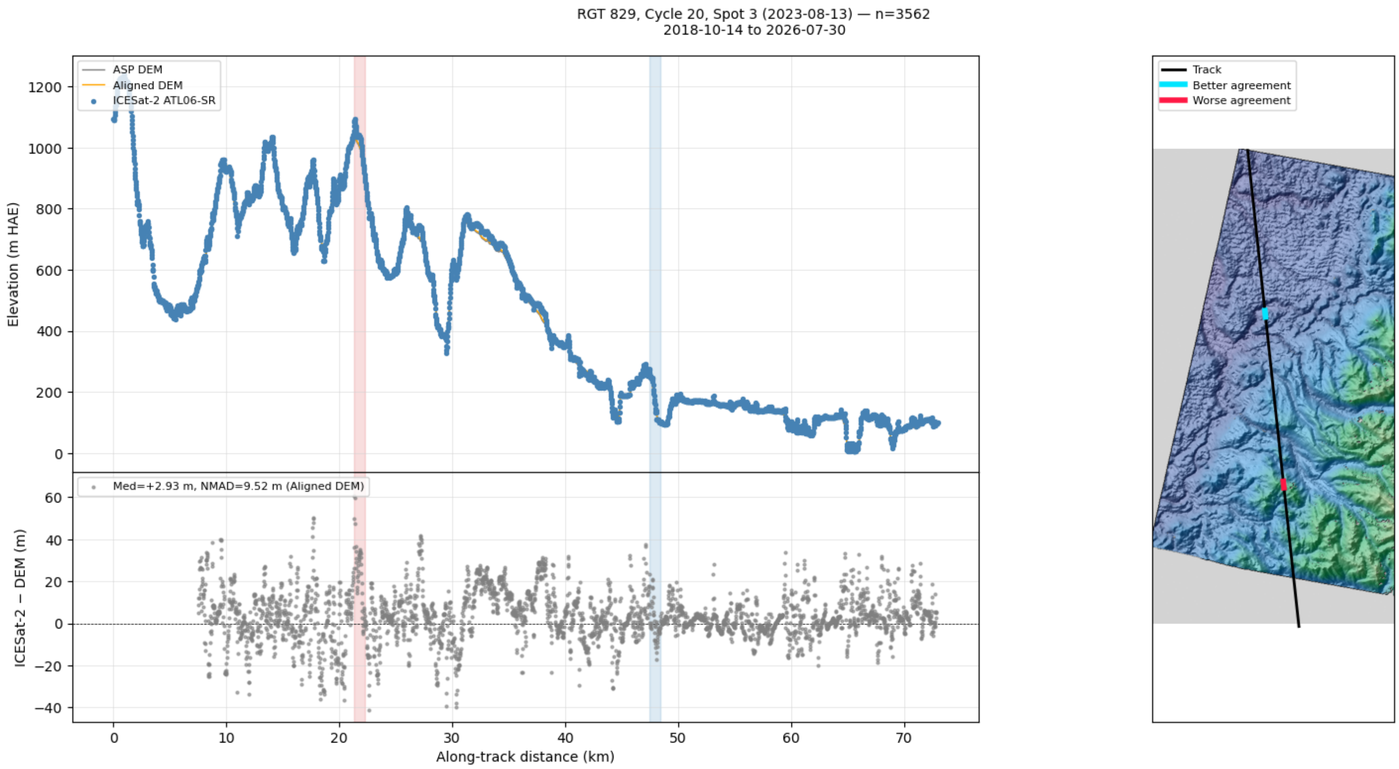


*Figure 13: Elevation profile along the ICESat-2 track after pc_align. The aligned DEM is overlaid on the profile and used to recompute the height differences shown in the lower panel.*

# ICESat-2 ATL06-SR Agreement Segments (Aligned DEM)

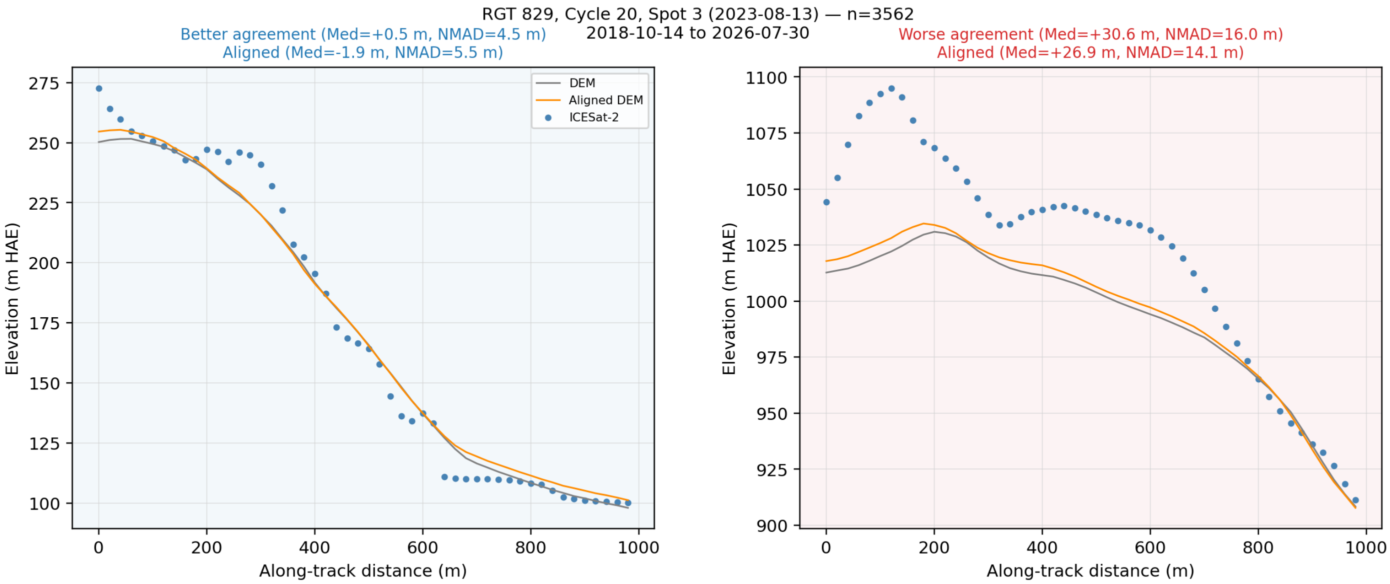


*Figure 14: The same better- and worse-agreement segments as above, now with the aligned DEM overlaid. Segment selection is held fixed so Median/NMAD can be compared directly.*